

Scroll. It's all *real*.¹

Ayush Yadav — software engineer · Cincinnati, Ohio

the capstone — agentic automl: seven gated phases —
case files: Applied · Glyph · Cadence
scroll — the story starts here
in a hurry — the work · the evidence · the resume

1. every claim on this page links to its evidence. ↩

01 / 07

This is a record of what I've built. *And how each piece was checked before I called it done.*

I'm a recent computer-science graduate from Miami University. I build the whole path — the interface, the services behind it, the data they run on, and the machine learning when the problem calls for it. The claims on this paper carry receipts.

ap·prov·al

/ ə'pru:vəl / · noun

1. the moment a human signs for a machine's work. 2. the step I leave to a person.

n.b. — every project below opens into its own case file, source, or live build.

¶ 03 / 07 · the path — field records · 12:06

Thousands of service tickets. Zero structure.

*This is where I learned that the work
starts in the mess.*

jun 2025 — may 2026
miami university · oxford, ohio

ITSM Data Integration Student Associate

jun 2025 —

The tickets arrive without shape. A million-plus service records, and no two name their problem alike.

1m+ operational records — high-volume python + sql transforms

the fix —

Python and SQL teach the records shape. The transforms feed OAS and Tableau, and the dashboards they fill are read across university departments.

oas/tableau reporting — dashboards across departments

the bot —

A chatbot in Slack takes the routine questions — wired to Workday, Oracle Analytics, and Tableau, with access governed at every joint.

governed data access — workday · oracle · tableau apis

the assistant —

A second assistant reads policy and answers with its sources cited — and drops the answers whose quotes it cannot verify.

19/20 cited-source sweep — policybot, the policy assistant

The agent drafts the whole pipeline.

Nothing runs until a human says go.

Agentic AutoML turns a dataset and a goal into a structured, auditable ML workflow. The agents plan it and run it, then stop at the seventh phase and wait for a person.

7 phases · every one gated · langgraph + mcp orchestration

what the run is held to – quoted from the case file

Require human approval before generated actions alter workflow state.

read the case file – agentic automl platform —

```
1.0 ingest
2.0 explore
3.0 preprocess
4.0 engineer
5.0 train
6.0 evaluate
    the human gate – go / no-go
7.0 deploy
```

fig. 4.0 – the 7-phase gated lifecycle, abridged.

```
run · model                                metrics          status
metrics withheld – the caption says why
038 · log-reg                               [REDACTED]        approved
```

fig. 4.1 – experiment registry, transcribed private-safe excerpt.
[REDACTED] – not published: the repository is private, and the demo-data metric trail has not shipped. see the case file.

Applied

Your inbox already knows where you applied.

Applied reads it – inbox in, a pipeline of real applications out.

macro-f1 0.98 – 96-sample gate
the case file —
last verified 2026-07

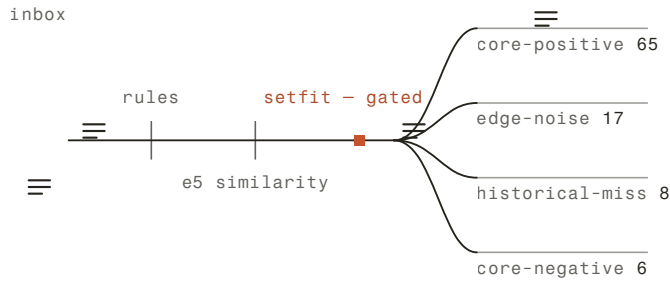


fig. 5.0 – the sorting line – the 96-message eval set through rules → e5 similarity → gated setfit.
96 messages · 12 per label × 8 labels · macro-f1 0.9791 – committed eval baseline · ships in-browser: int8 onnx, 22.8 mb

Glyph

No frameworks. Just math, memory, and C++.

Then OpenMP and SIMD together made it 3.5× faster.

openmp+simd dot kernel vs -O3 baseline, benchmarks
the case file →
last verified 2026-07

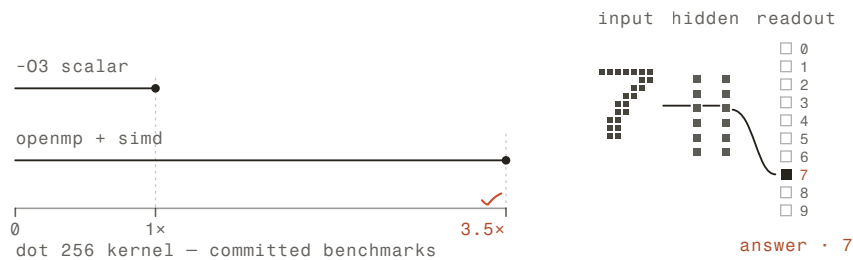


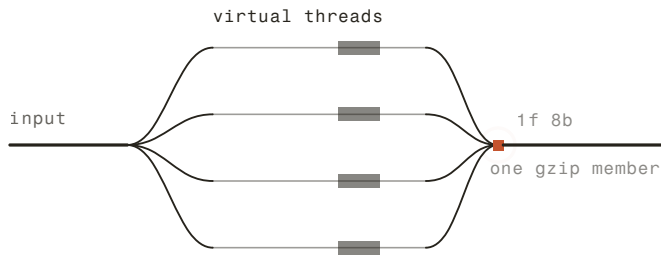
fig. 5.1 – the race – equal time, measured distance · the forward pass – a drawn 7, answered.
3.5× – openmp+simd dot kernel vs -O3 baseline (dot 256), committed benchmarks · avx-512 · avx2 · neon · wasm-simd128

jetpack-compress[↗] (<https://jetpack-compress.vercel.app>)

One gzip stream. Every core writing it.

jetpack-compress splits the bytes across virtual threads – and stitches one byte-valid member.

@ jetpack-compress/blob/af2c4b1/README.md
the live demo ↗ (<https://jetpack-compress.vercel.app>)
source ↗ (<https://github.com/jetpack-compress>)
system card ↗ (<https://jetpack-compress.vercel.app/system-card>)
last verified 2026-07



gzip this figure in your browser →

fig. 5.2 – split → parallel → stitch – virtual-thread gzip
 into one byte-valid member.
 ~6.5× vs single-threaded java.util.zip (±50%, quick benchmark)
 · 72 tests, 0 failures · jdk 25 @ af2c4b1

Cadence

A calendar and task app you drive in plain English.

Type a sentence and it files the event or task – and now schedules Google Meet meetings with multi-attendee invites.

frontend + 511 backend, vitest
 the case file →
 last verified 2026-07

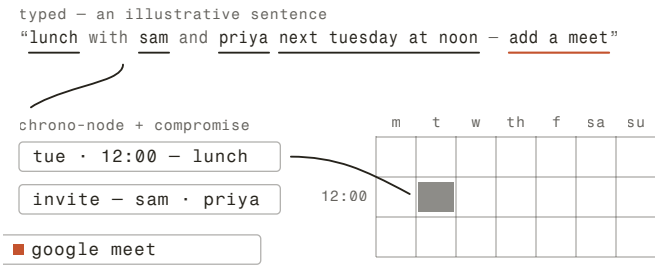


fig. 5.3 – the parse – a plain-english sentence resolves into chips on the week grid + a google meet link.
 chrono-node + compromise · google meet – multi-attendee invites · 1,145 tests passing – taskflow-calendar @ 69a59e7

cited above, argued in full –
 master inventory pipeline · policybot

also live, without a case file –
 lifequest ↗ (https://getlifequest.vercel.app) – a social-good concept for job-seekers; a playable prototype · system card ↗ (https://getlifequest.vercel.app/system-card)

¶ 06 / 07 · how i work — after dark · 21:07

Make it learn.

271 backend tests — jobtracker validation ledger

Make it fast.

openmp+simd dot kernel — 3.5× over -O3, committed benchmarks

Make it honest.

19/20 cited-source sweep — policybot validation ledger

fig. 6.0 — the litany, with receipts.

gate	disposition
refused — the gate stopped the run	
classifier gate — jobtracker	✓ passed
cited-source sweep — policybot	— refused
benchmark suite — fast-mnist	— refused

fig. 6.1 — the gates, kept.
passed — human signed off

¶ 07 / 07 · the approval gate — closing page · 22:41

Every pipeline I build ends with a
human decision.

This one ends with yours.

Ayush Yadav

availability — open to new-grad software, data, and ml engineering roles

cincinnati, ohio · it's 9:36 pm here right now — the 🕒 clocks are the day this paper records

aesh_1055@icloud.com [resume](#) [github](#) ➤ (<https://github.com/yadava5>)

[linkedin](https://www.linkedin.com/in/ayush-yadav-developer/) ➤ (<https://www.linkedin.com/in/ayush-yadav-developer/>)

Email me — I read everything.



references — footnote 1, kept.

1. macro-f1 0.9791, 96-sample gate	jobtracker — receipt 05 —
2. 3.5× — openmp+simd dot kernel vs -O3 baseline	fast-mnist-nn — receipt 02 —
3. 1,145 automated tests — 634 frontend + 511 backend	taskflow-calendar — receipt 01 —
4. 72 tests, 0 failures on jdk 25 jetpack-compress — readme @ af2c4b1 ➤ (https://github.com/yadava5/jetpack-compress/blob/af2c4b1/README.md)	
5. 19/20 cited-source sweep	policybot — receipt 01 —
6. 1M+ operational records transformed	master-inventory — receipt 05 —
7. 7-phase gated lifecycle, human go/no-go	automl — the case file —

on file — references, the other kind.

“...Ayush showed strong judgment, independence, and follow through. He understood intent, not just requirements. He took feedback, iterated quickly, and delivered usable outcomes. He is disciplined, thoughtful, and reliable. Ayush would add value to any data, analytics, or platform team that expects real contribution and accountability....”

Randall Vollen ✎ (<https://www.linkedin.com/in/randall-v-30745a4/>) — Hands on Experienced Data, AI & Analytics Executive managed the data internship, 2025–26 · recorded 2026–01

excerpted, not edited — the full recommendation is on the linkedin profile in the colophon.

the evidence index — every claim on file —

07 / 07

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set in fraunces, newsreader & fragment mono

two inks. one line. seven chapters.